

Composite Information Server

Businesses are constantly changing. Mergers, new products, new markets, and compliance mandates are but a few of the major drivers in a dynamic company. Information is the life blood that helps a company manage change and succeed.

Composite Software's data services solution delivers existing data to your next-generation applications. Data services are a critical component in any information infrastructure or service-oriented architecture (SOA). Composite's data services solution, based on our award winning Enterprise Information Integration (EII) technology, enables you to:

- **Respond faster to business requests** by providing your new and evolving applications with the data they require faster than ever before.
- **Reduce costs** by reusing not only your existing data in new and powerful ways, but also reusing data services across projects.
- **Minimize SOA risks**, protecting performance and ensuring security, as you SOA-enable existing systems.

PRODUCT

The Composite Information Server (CIS) is a multi-threaded, 100-percent Java application that non-invasively accesses existing data from disparate sources, then combines, abstracts and simplifies the complex data, and delivers it via data services or relational views. CIS features patent-pending IQ (Intelligent Query) optimization technology, which delivers the highest performance in the industry. CIS provides caching capabilities, a metadata repository, and supports multiple methods of handling data security. CIS also includes the Composite Studio, a GUI-based tool for data modeling, data services building, services management, and security configuration.

ADVANTAGES

The Composite Software solution for data services provides many advantages, including:

Leverages existing data assets

A data services layer lets you efficiently and effectively reuse your data across many new applications.

Accelerates new development projects

A comprehensive data services solution lets you quickly and easily develop needed data services or relational views.

Simplifies SOA migration

A practical data services solution that supports phased adoption across both traditional and SOA architectures.

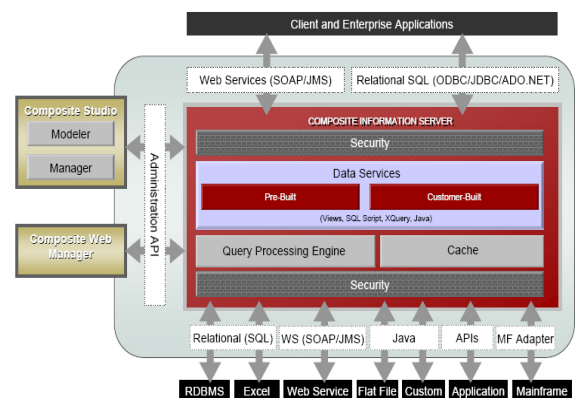


Figure 1 - Product Architecture

Composite Information Server

FEATURES

Query Processing – Composite's run-time environment for running and optimizing queries.

- Query optimization – The most efficient and cost-effective queries are executed from multiple query plans.
- SQL Script – Implement stored procedures with Composite's own scripting language.
- Multiple join algorithms – Query options for added flexibility.
- SQL 99 – Full support of the latest SQL standards.
- Alerts – resource, event and user-defined triggers. Published API to handle custom java alerts.
- Scheduling – Execute queries based on set times.
- Results email – Receive query results via email.

Modeling and Metadata Management – Composite allows powerful modeling of information views with easy-to-use graphical tools.

- Bottom-up modeling – Design views by combining data from disparate systems without worrying about underlying source access and format complexity.
- Top-down modeling – Create unbounded views without first needing to define data sources.
- Public metadata API – Available as a collection of Web services and built-in procedures.
- XML shaping – Transform data from a tabular to hierarchical structure and vice versa. Access XML data with SQL.
- XPath transformation – Establish arbitrary complex mapping of XML Schema elements into XML output
- Views dependency graph – Graphically display what data sources views are dependent on and vice versa.
- Logic modeling – Create views by using scripting languages for added convenience and flexibility.

Version Control - Manage development and changes in a controlled manner.

- Version-based Import and Export - Save and restore resources from directories and files.
- Resource Locking - Protection against inadvertent modifications and overwrites.
- Change History - Track changes made by users with annotations.

Metadata Repository – Composite's repository for storing and maintaining the metadata of all resources, such as data sources, views, and procedures.

- Import and export – Upload and download external resources with various formats.
- Schema storage and viewing – View stored schemas.
- Transformation – Convert XML data from/to tabular form.
- Schema change notification – Receive notice when schemas upon schema changes.
- Metadata exposure – Access metadata of physical data source.
- Open API – Web services-based for easy access and sharing.
- Third-party compatibility – Import and export metadata with ERStudio and Erwin.

Caching – Composite allows views that are repeatedly used to be stored locally for improved performance. Supports caching in local database or files.

- Event driven refresh – Refresh views based on defined business criteria.
- Scheduled refresh – Refresh views based on set times.
- Manual refresh – Refresh on demand as needed.
- Hybrid memory/disk usage – Balance usage of memory versus disk space for optimal performance.
- Parameterized caching – Caching of parameterized procedures and Web services calls.

Data Source Management – Composite supports numerous data sources and provides tools to manage their discovery, connection, and operation.

- Multiple data source handling – Supports JDBC/ODBC databases, SOAP/WSDL, LDAP, HTML, flat files.
- Introspection – Probe physical data sources and select desired resources.
- Connection pool sharing – Share access to data source to avoid bottleneck.
- Native XML support – Supports XML internally for fast parsing and joins.

Scalability and Reliability – Composite ensures scalability and up-time for enterprise-class applications.

- Clustering – Link multiple servers to handle variable load.
- Listener process – Continuous pulse monitoring for maximum availability and up-time.

Security – Composite enforces security throughout the system, from the moment queries are initiated.

- User authentication – One login and password permits access to all appropriate data sources.
- Database access – Table access could be restricted down to column and row level. Data access could also be restricted to module and function level.
- SSL over HTTP – Certificate based authentication and WSS authentication support.

Administration – Native system management and administration capabilities accessible through API.

- Real-time system indicators – Monitor key system metrics and tune for optimal performance.
- Browser-based console – Measure and configure the server from a light-weight web console.

SPECIFICATIONS

Data Sources

- IBM DB2 v7, v8
- Informix 9.x
- MS SQL Server 2000, 2005
- MySQL 3.x, 4.x
- Netezza NPS 2.x and 3.x
- Oracle 8i, 9i, 10g, RAC
- Sybase 12.5
- Teradata v2 r6.0, v2 r6.1, v2 r6.2
- Any ODBC/JDBC source
- MS Access, and Excel
- LDAP v3
- XML (Flat files or via HTTP)
- TIBCO EMS, Sonic MQ
- Flat Character Delimited Files

Application Sources

(See [Composite Application Data Services data sheet](#))

- SAP 4.6, 4.7, 5.0, 6.0
- Siebel eBusiness Applications 7.x
- Oracle E-Business Suite 11.5.8, 11.5.9, 11.5.10
- PeopleSoft 8.47.12 or later
- Salesforce.com

Access Interfaces

- SOAP 1.1, JDBC 2.0 & 3.0, ODBC 2.x
- SOAP/JMS - TIBCO EMS, Sonic MQ on AXIS/AXIS2
- ADO.NET 32bit and 64 bit

Directory Services

- Windows Server 2003 Active Directory
- iPlanet 4.1.6 sp1
- Composite

Repositories

- MySQL 3.x, 4.x
- Oracle 8i, 9i, 10g
- Sybase 12.5

Standards

- JDK 1.5, J2EE 1.3, JAAS, JNDI
- WSDL 1.1, SOAP 1.1, XSLT 1.0, XML Schema 1.0, SAX and DOM
- SQL 92, 99
- Unicode support

Platforms

- 32-bit, 64-bit JVM
- Server
 - Microsoft Windows 2000, 2003, XP
 - SUN OS 5.8 / Solaris 9 or higher
 - Red Hat Enterprise Linux AS 3, 4, 32/64 bit
 - SUSE Linux Enterprise Server v9.3, v10, 32/64 bit
 - IBM AIX 5.2, 52000-04
 - HP-UX 11.11, B.11.11.0412.5 Gold Base 11i
- Client for Composite Studio

COMPOSITE SOFTWARE

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